

TANISHQ SHUKLA

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EDUCATION

Bachelor of Technology | Chemical Engineering | **Indian Institute of Technology (BHU), Varanasi** 2021 - 2025

SKILLS

Languages : C/C++ , Typescript, Php, Python, Rust, HTML, CSS, SQL, Golang

Technology/Frameworks : MERN, AWS, EC2, S3, Redis, Nginx, FastAPI, Postgresql, Docker, Grafana, Langchain, Langsmith

Soft Skills : Leadership, Team Management, Problem Solving, Critical Thinking, Communication, Time Management

Interests : Software Development, Microservices, LLMs, RAG, Agentic System Design and Frameworks

WORK EXPERIENCE

Founding Engineer | Surgegrowth

Feb'25 - Present

Bengaluru, Karnataka (Onsite)

- Engineered an MCP server (FastMCP) serving ~12k tool calls/day across 9 task-oriented tools for Claude.ai/Cursor, secured with a full OAuth 2.1 flow—PKCE, dynamic client registration, and rotating refresh tokens.
- Co-built billing & usage-based pricing via Dodo—idempotent payments processing \$10k+/mo at 99.99% success, subscriptions, and admin-configured rate cards metering usage across 40+ orgs with prepaid credit lots (FEFO).
- Replaced paid scraping APIs with in-house Facebook/Instagram & Google Ads scrapers and a YouTube downloader, leveraging Decodo residential proxies as egress IPs to bypass anti-bots—cutting the bill \$1,800→\$200/mo (~89%).
- Co-designed distributed job infrastructure handling ~50k requests/day (30–50/sec peaks)—APScheduler cron on a Postgres job store, Cloud Pub/Sub dispatch, and Cloud Run Jobs/Functions with atomic claims for exactly-once runs.
- Developed a media-processing engine (FFmpeg) processing 500k+ frames/day (~1k req/day), with self-hosted ML inference (MDX-Net/ONNX, DeepFilterNet) for voice-enhancement—replacing json2video (\$50/mo saved).
- Built a multi-model media generation gateway integrating Veo 3.1, Nano Banana & Seedance APIs, with per-provider rate-limit handling and async queues to balance request load across providers.
- Instrumented per-org usage analytics on ClickHouse for sub-second queries across AI tokens, 3rd-party API costs & workflows, synced from the credit ledger via PeerDB CDC to power real-time usage & cost dashboards.

SDE Intern | Kimcc

May '24 - July '24

Bengaluru, Karnataka (Onsite)

- Led two key projects, Pulse & Time Diary, within a week of joining; revamped Pulse analytics, lifting engagement 25%.
- Engineered a cross-platform Electron desktop app with secure auth, offline-first sync, and reliable productivity tracking.
- Resolved critical production bugs and built complex MongoDB aggregation pipelines that cut query time ~30%.
- Automated recurring workflows with Node.js cron jobs and scripts to accelerate task execution and boost productivity.

SDE Intern | Oton

Dec '23 - Feb'24

Bengaluru, Karnataka (Remote)

- Transformed workflow with a custom Slack bot (Node.js) automating critical processes and improving team efficiency.
- Utilized Prisma to handle complex queries and administer PostgreSQL at scale, cutting query response time 25%.
- Crafted cursor-based pagination and integrated Jira APIs for boards, sprints, issues & tasks, enhancing dashboards.
- Implemented Redis Bull queues for efficient batch processing, smoothing system load under bursty job volumes.

PROJECTS

Sentinel | Agent Evaluation & Red-Teaming Platform | [Github link](#)

- Built an agent evaluation platform with pluggable check profiles (groundedness, answer-relevance, refusal, PII-leak) and per-agent thresholds, auto-generating test cases from each agent's KB and scoring via calibrated LLM judges.
- Designed an OWASP-aligned red-team harness with a 10+ attack catalog across 5 categories (prompt injection, jailbreak, PII exfiltration, system-prompt leak, instruction override) and canary-token & LLM-judge detection.
- Engineered a hybrid RAG pipeline (pgvector dense + ParadeDB BM25, RRF-fused) with per-agent KB isolation and defense layers (spotlighting, PII egress filtering) that cut attack success rate from 100% to under 20%.
- Shipped a full-stack agent assurance dashboard (FastAPI + Next.js + LangGraph) with span-level trace inspection, a cross-agent leaderboard, and live analytics—deployed via Docker Compose with async jobs on Redis/RQ.

OutageRadar | Distributed RSS Feed Monitor | [Github link](#)

- Built a production-grade async RSS status tracker monitoring OpenAI, Claude feeds via an event-driven Redis pipeline.
- Architected producer/consumer flow on Redis with atomic SETNX claims for idempotent processing across workers.
- Engineered aiohttp fetcher with conditional GET (ETag), adaptive polling, Strategy/Registry parsers for new providers.
- Deployed via Docker Compose with AOF-persisted Redis, health checks, structured JSON logs, and graceful SIGTERM shutdown.

LEADERSHIP EXPERIENCE

Startup Assistance Program Manager | E-Cell IIT BHU

May'23 - May'24

- Spearheaded the Startup Assistance Program (SAP), acting as the principal liaison for 20+ startups at E-Summit'24.
- Promoted IIT BHU's entrepreneurial ecosystem, boosting E-Cell visibility, through participation at IIT Kanpur.
- Facilitated critical engagements for 15+ startups, ensuring successful and impactful presence at E-Summit'24.